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Diffusion-ordered spectroscopy NMR DOSY: an all-in-one tool to simultaneously follow side reactions, livingness and molar masses of polymethylmethacrylate by nitroxide mediated polymerization†

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In this work, we report the use of the diffusion-ordered spectroscopy (NMR DOSY) nuclear magnetic resonance (NMR) technique to follow the nitroxide-mediated polymerization (NMP) of methyl methacrylate (MMA), well known as a difficult monomer to polymerize by NMP. Two syntheses were carried out either in the presence or absence of a small amount of styrene at 80 °C, using *N-tert*-butyl-*N*-(1-diethylphosphono-2,2-dimethylpropyl)nitroxide (SG1) as a control agent and 2-methylaminoxypropionic-SG1 alkoxyamine (BlocBuilder) as the initiator. Diffusion coefficients determined from NMR DOSY experiments could be accurately correlated to the weight average molar masses M_w of PMMA, and could thereby be used to confirm that the polymerization of pure MMA by NMP is uncontrolled due to disproportionation side reactions leading to non-reversible termination (*i.e.* dead chains). In a similar manner, the livingness of the PMMA chains grown in the presence of a small amount of styrene was confirmed by the linear evolution of the diffusion coefficient *versus* time (conversion) and the retention of the alkoxyamine chain end. Livingness was further assessed by NMR DOSY analysis of chain extensions with *n*-butyl acrylate. This all-in-one technique permits disproportionation side reactions, livingness, and evolution of the molar masses with time to be simultaneously followed.

Introduction

A wide range of monomers, including styrene, acrylates, acrylamides, acrylonitrile and many others, can be successfully polymerized *via* nitroxide-mediated polymerization (NMP),[1–3] allowing the design of complex macromolecular architectures. However, the polymerization of methacrylic esters *via* NMP presented a great challenge for a long time due to their high activation–deactivation equilibrium constant <*K*>, which leads to a high proportion of propagating macro radicals and encourages irreversible termination events.[4–6]

In 2005, Charleux *et al.*[7,8] presented a theoretical approach to describe the copolymerization kinetics in controlled/living copolymerization systems operating *via* reversible termination. They theoretically determined the variation of the average activation–deactivation equilibrium constant <*K*> of the copolymerization and showed that the addition of less than 10 mol% of styrene S to methyl methacrylate (MMA) (2.2–8.8 mol% based on the monomers) was enough to achieve control of the NMP of MMA at 90 °C up to high conversions by sufficiently decreasing the value of <*K*>. In 2010, the successful SG1-mediated polymerization of MMA was also reported under similar experimental conditions, by replacing styrene by acrylonitrile AN as a comonomer.[9] P(MMA-*co*-AN)-SG1 was then used as a macro-initiator to obtain block copolymers with *n*-butyl acrylate (BA) and styrene (S) by chain extension, confirming their living character. The percentage of BA required as a comonomer to perform successful SG1-NMP synthesis of PMMA was much higher than for styrene or acrylonitrile; 75% of BA was required to reach a <*K*> value comparable to that of the styrene comonomer system.[10] All these reported studies were based on theoretical approaches and mainly used ^{1}H-nuclear magnetic resonance (NMR) spectroscopy and size

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exclusion chromatography (SEC) to characterize the livingness of the NMP polymerizations. ^{31}P NMR has also been used to assess chain end activity since SG1 contains a phosphorus atom.[8,9]

The NMR DOSY technique was developed as a general method for the identification of different components in mixtures. It is based on a pulse field gradient spin-echo NMR experiment accompanied by diffusions of different species.[11] Two-dimensional spectra are obtained which correlate the observed diffusion coefficients of each component with their corresponding chemical shifts.[11–14] During the last decade, the use of this NMR technique has experienced a real explosion in many fields, particularly for medical, pharmaceutical and biomedical analysis,[15–17] food analyses,[18,19] and natural and synthetic complex analyses.[20–22]

In polymer science, NMR DOSY has also received considerable attention since the values of the determined diffusion coefficients reflect the effective size of macromolecular species and depend on the physical properties of their surrounding environment in solution.[23–26] This technique is generally considered as a chromatographic method for physical component separation, since it allows the estimation of polymer weight-average molar mass, but without requiring any particular sample preparation or time analysis in contrast to SEC which is time consuming, requires large quantities of solvent, and prior sample preparation.[23–26] Moreover, Delsuc *et al.*[23] reported the use of NMR DOSY for the accurate estimation of dispersity of linear polymers by performing simple diffusion experiments. In 2012, Li *et al.*[24] reported the first accurate determination of weight-average molar mass M_w *via* NMR DOSY for polymers obtained by atom transfer radical polymerization (ATRP), reversible addition-fragmentation chain transfer (RAFT), and ring-opening metathesis polymerization (ROMP). Also, this same technique was considered as a tool to check block copolymer formation.[25,26]

In this work, we focus on the all-in-one NMR DOSY technique to demonstrate its efficiency and versatility in following the mechanism of polymerization regarding its livingness *via* molar mass evolution in homopolymerisation and chain extension.

Experimental section

Materials

Methyl methacrylate (MMA, Aldrich, 99%), *n*-butyl acrylate (BA, Aldrich, 99%) and styrene (S, Aldrich, 99%) were used as received. The alkoxyamine BlocBuilder® (2-methylaminoxypropionic-SG1) and the nitroxide SG1 (88%) provided by Arkema Chemicals were used as received. PMMA and PS standards for NMR DOSY calibration were provided by Polymer Laboratories. All solvents were used without further purification.

Macromolecular synthesis

PMMA-SG1. In a round-bottom flask, MMA (26.8 g, 0.268 mol) was mixed with the BlocBuilder® alkoxyamine MAMA (0.510 g, $0.134.10^{-2}$ mol) as an initiator ([MMA]/[MAMA] = 200), and SG1 (0.0394 g, $0.134.10^{-3}$ mol) ([SG1]/[MAMA] = 0.1). After degassing for 30 min, the flask was placed into an oil bath thermostated at 80 °C and the reaction was stopped after 4 h.

Poly(MMA-*co*-S)-SG1. In a round flask, MMA was mixed (25 g, 0.25 mol) with Styrene (2.5 g, 0.024 mol), BlocBuilder® alkoxyamine MAMA (0.521 g, 0.137×10^{-2} mol) and SG1 (0.0405 g, 0.137×10^{-3} mol) ([MMA-S]/[MAMA] = 200, [SG1]/[MAMA] = 0.1). As for the first solution, this mixture was degassed for 30 min, then placed into an oil bath thermostated at 80 °C, and was stopped after 4 h.

Poly(MMA-*b*-BA). In a round flask, a mixture of BA (5.68 g, 0.0444 mol), PMMA macro-initiator (1.5 g, 1.04×10^{-4} mol) and SG1 (0.0039 g, 1.33×10^{-5} mol) was first degassed for 30 min, and then placed into an oil bath thermostated at 120 °C. The reaction was stopped at 240 min, [BA]/[PMMA] ≈ 430, [SG1]/[PMMA] ≈ 0.12.

Poly((MMA-*co*-S)-*b*-BA). In a round flask, BA was mixed (8.05 g, 0.0628 mol) with a P(MMA-*co*-S) macro-initiator (1.5 g, 1.10×10^{-4} mol) and SG1 (0.0055 g, 1.88×10^{-5} mol). As for all other syntheses, the mixture was degassed for 30 min then placed into an oil bath thermostated at 120 °C for 240 min. [BA]/[P(MMA-*co*-S)] ≈ 500, [SG1]/[P(MMA-*co*-S)] ≈ 0.16. For all the polymerizations, samples were withdrawn during the polymerization for further monomer conversion by ^{1}H-NMR analyses (Fig. SI.1†). All these samples were precipitated twice into a methanol (90/10) mixture for SEC and NMR DOSY analysis. The final products were also precipitated in the same mixture, filtered and dried in a vacuum oven until constant weights were reached.

Characterization techniques

Size exclusion chromatography. The crude polymer solutions were analyzed by SEC at 30 °C with THF as an eluent at a flow rate of 1 mL min^{-1}. The SEC system was equipped with three Waters Styragel columns HR 0.5, 2, 4 working in series (separation range 1×10^2–3×10^6 g mol^{-1}) and a Viscotek ERC 7515-A refractive index detector. Apparent weight-average molar mass M_w and dispersity $Đ$ (M_w/M_n) were calculated with a calibration derived from PS standards using OmniSEC-software. All polymers samples were prepared at 1 to 5 g L^{-1} concentrations and filtered through PVDF 0.45 mm filters.

NMR experiments. All NMR experiments were performed at 25 °C on a Bruker Avance 400 spectrometer equipped with a Bruker 5 mm BBFO probe and a gradient amplifier, which provides a *z*-direction gradient strength of up to 47.5 G cm^{-1}. The temperature was maintained constant within ±0.1 °C by means of the BCU 05 unit. All NMR DOSY experiments were performed using the bipolar longitudinal eddy current delay pulse sequence (BPLED). The spoil gradients were also applied at the diffusion period and the eddy current delay. Typically, a value of 2 ms was used for the gradient duration (δ) 150 ms for the diffusion time (Δ), and the gradient strength (g) was varied from 1.67 G cm^{-1} to 31.88 G cm^{-1} in 32 steps. Each parameter was chosen to obtain 95% signal attenuation for the slowest diffusion species at the last step experiment. The pulse repetition delay (including acquisition time) between each scan was larger than 2 s. Data acquisition and analysis were performed using the Bruker Topspin software (version 2.1). The T1/T2 analysis module of Topspin was used to calculate the diffusion coefficients and to create two-dimensional spectra with NMR chemical shifts along one dimension and the calculated diffusion coefficients along the other.

The diffusion measurement was carried out by observing the attenuation of the ^{1}H NMR signals during a pulsed field gradient experiment. In NMR DOSY experiments using the Bipolar Longitudinal Eddy current Delay (BPLED) pulse sequence,[27] this intensity change is described by eqn (1):

$$I = I_0 e^{-D\gamma^2 g^2 \delta^2}(\Delta^{-\delta}/3) \tag{1}$$

where I is the observed intensity, I_0 is the reference intensity (unattenuated signal intensity), D is the diffusion coefficient, γ is the gyromagnetic ratio of the observed nucleus, g is the gradient strength, δ is the length of the gradient, and Δ is the diffusion time.

NMR DOSY provides a means for "virtual separation" of compounds, by providing a 2D map in which one axis is the chemical shift while the other is the diffusion coefficient.

Here, we can mention that molar masses are apparent molar masses and were obtained from both SEC (using PS calibration) and NMR DOSY spectroscopy (using PMMA and PS calibrations).

Results and discussion

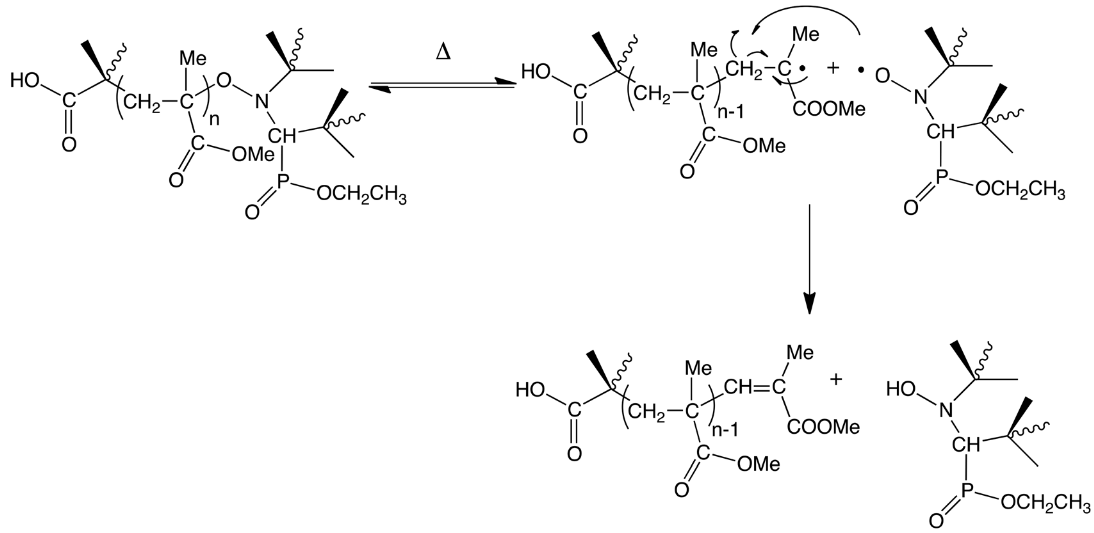
Two different behaviors are observed when studying the variation of the logarithmic monomer concentration *versus* time and the evolution of $M_{n,SEC}$ and $Đ$ *versus* conversion of MMA in either the presence or the absence of a small amount of styrene (Fig. SI.1†). Indeed, when adding 10 mol% of styrene, the PMMA exhibits features of controlled polymerization systems. The variation of the logarithmic monomer concentration *versus* time shows a fast initiation followed by a linear evolution, the experimental molar masses M_n increase linearly *versus* conversion, and low dispersity values $Đ$ (*i.e.* below 1.3) are obtained. In contrast, conversion in the pure MMA polymerization reaches a plateau within the first 10 minutes (corresponding to 28% of monomer conversion), which is in accordance with the work of Charleux *et al.*[7] No significant change in the molar masses *versus* conversion occurs (Fig. SI.2† and Table 1). The SEC distributions of the polymers at the first and the last time points in the pure PMMA system indeed illustrate constancy of the molar mass, whereas a shift to higher molar masses is observed in SEC chromatograms of the P(MMA-*co*-S) system, without any observed shoulder (Fig. SI.3†).

All these observations confirm the lack of control of PMMA synthesis using SG1-NMP in the absence of styrene,[7,8] explained by the development of side reactions such as irreversible termination between the propagating radicals and/or disproportionation reactions *via* hydrogen transfer from propagating radicals to free-SG1 nitroxide (Scheme 1).[6,28]

Table 1 Kinetics and macromolecular characteristics of PMMA and P(MMA-*co*-S) macro-initiators and their diblock copolymers PMMA-*b*-PBA and P(MMA-*co*-S)-*b*-PBA by ^{1}H-NMR, SEC and NMR DOSY

Sample		NMR	SEC		NMR DOSY		
						M_w [b] (g mol^{-1})	
Name	Time (min)	Conversion %	M_w (g mol^{-1})	$Đ$	$D \times 10^{10}$ [a] (m^2 s^{-1})	PMMA	PS
PMMA	10	28.8	13 800	1.60	1.13	15 500	15 800
	60	27.9	14 600	1.43	1.14	15 200	15 200
	120	27.9	15 000	1.40	1.15	15 000	15 000
	150	27.7	13 500	1.63	1.15	15 000	15 000
	180	28.8	13 500	1.56	1.14	15 200	15 200
	240	28.3	16 800	1.30	1.08	16 900	16 900
P(MMA-*co*-S)	10	7.5	3900	1.68	2.28	4680	4100
	60	15.7	6900	1.26	1.90	6430	5900
	120	22.7	8500	1.26	1.45	10 300	9700
	150	26.7	9500	1.23	1.37	11 300	10 800
	180	30.8	10 300	1.24	1.32	12 000	11 600
	240	36.7	12 600	1.19	1.16	15 200	14 700
P(MMA-*b*-BA)	30	0	16 400	1.34	1.10	16 600	16 600
	60	0	16 400	1.35	1.09	16 800	16 800
	150	0	16 750	1.34	1.12	16 200	16 200
	240	1.4	18 300	1.29	1.09	16 800	16 800
P(MMA-*co*-S)-*b*-PBA	30	7.5	15 300	1.18	1.13	15 800	15 800
	60	12.2	17 700	1.18	1.05	17 900	17 800
	150	30.4	25 000	1.20	0.87	25 000	25 300
	240	42.8	32 900	1.18	0.76	31 000	32 600

[a] Measured from DOSY NMR spectra. [b] Calculated from the PMMA (left) and PS (right) calibration curves in Fig. 2.



Scheme 1 Mechanisms of dissociation of alkoxyamine and the disproportionation reaction *via* hydrogen transfer to nitroxide SG1 leading to dead chains.

Consequently, a significant population of active macromolecular chain ends (*i.e.* alkoxyamine terminii) was only expected in the P(MMA-*co*-S) copolymer.[8]

Since ^{31}P NMR is able to detect SG1 end chains,[29] we used the MMA polymerization by NMP, as a model reaction to illustrate the ability of the all-in-one ^{1}H NMR DOSY technique, to follow the mechanism of polymerization, the livingness *via* molar masses evolution and chain extension to synthesize block copolymers from pure PMMA and P(MMA-*co*-S) polymers as macro-initiators, simultaneously. The study of the variation of the logarithmic monomer concentrations *vs.* time and the variations of M_n and $Đ$ *vs.* monomer conversions of these two systems shows that the features of controlled synthesis – linear variation of Ln$[M_0]/[M]$ with time and linear increase of M_n *vs.* conversion – were only observed in the P(MMA-*co*-S)-*b*-PBA system. The corresponding dispersity values $Đ$ are ≤1.2, and the monomer conversion reached 43% after 4 h (Fig. SI.4†). In contrast, the P(MMA-*b*-BA) system reached only 1.4% of BA conversion after 4 h, and no significant increase in molar mass occurred due to the high proportion of dead chains generated during the MMA polymerization. Table 1 summarizes the SEC and ^{1}H-NMR characteristics for the PMMA and P(MMA-*co*-S) macro-initiators and their corresponding chain extended polymers PMMA-*b*-PBA and P(MMA-*co*-S)-*b*-PBA.

Side reactions and livingness

In order to provide further insight into the chain extension processes, ^{1}H and DOSY NMR spectra of both PMMA and P(MMA-*co*-S) macro-initiators were recorded to highlight the difference in control at the final stage of the polymerization (Fig. 1).

The ^{1}H NMR spectra of PMMA and P(MMA-*co*-S) are shown in Fig. 1 (left). In the P(MMA-*co*-S) copolymer spectrum, two regions demonstrate the presence of styrene units: aromatic protons at 6.8–7.3 ppm, and PMMA methoxy protons in 2.2–3.7 ppm corresponding to MMA-centered triads with MMA–MMA–MMA (3.3–3.7 ppm), MMA–MMA-S (2.9 ppm) and S–MA–S (2.4–2.5 ppm).[29–31] Due to the low amount of S, MMA–MMA–MMA and MMA–MMA-S triads predominate in this region. In the 0.5–0.6 ppm region, a shoulder is also detected and corresponds to the methyl groups of the SG1 nitroxide end-chains. This peak is not observed in the spectrum of PMMA, confirming the expectation that the reactivatable function is only present in the P(MMA-*co*-S) copolymer.

The zone containing key information on the PMMA microstructure and therefore control of the NMP process is between 5 and 6.5 ppm. No significant signal is observed for the P(MMA-*co*-S), whereas signals characteristic of a methacrylic function are evident in the PMMA spectrum (and are not explained by residual MMA since extensive purification was performed). This function arises from irreversible termination by disproportionation, *i.e.* hydrogen transfer from a PMMA propagating radical to free-SG1 nitroxide, giving dead chains which are unable to propagate (Scheme 1). The polymerization therefore behaves more as a classical free radical polymerization.

So, the aim of this work is, for the first time, the use of the diffusion ordered spectroscopy technique NMR DOSY as a powerful tool to confirm (1) the origin of these signals and (2) the importance of styrene for side reaction elimination and consequently the control of PMMA by NMP.

The presence of this methacrylic function (Fig. 1 top left) corresponds to the presence of hydrogen on unsaturated carbon covalently linked to the macromolecular chain. Indeed, the signals display the same diffusion coefficient as the PMMA, highlighting the formation of unsaturated bonds during the NMP process of the MMA monomer (Fig. 1 top right). Such a phenomenon is clearly not observed for P(MMA-*co*-S) for which neither ^{1}H nor NMR DOSY show the presence of an unsaturated bond (Fig. 1 bottom left & right).

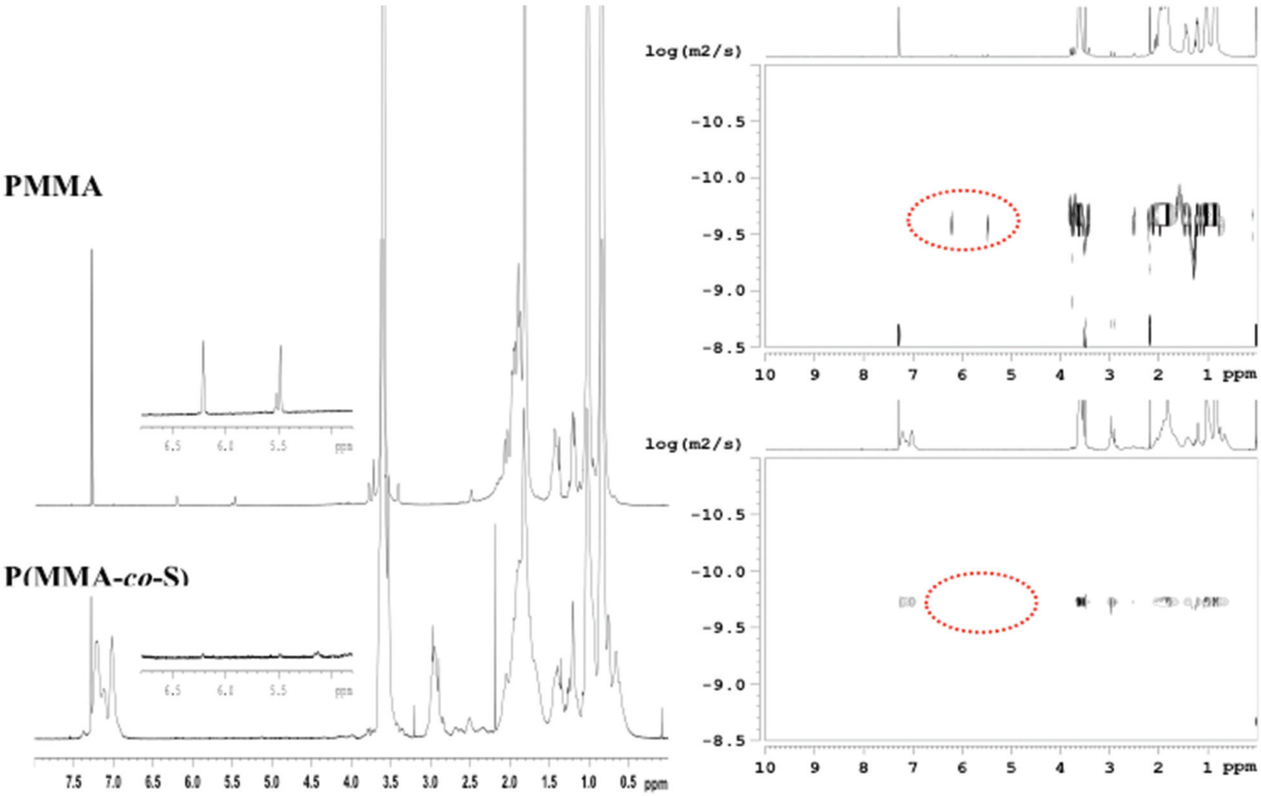


Fig. 1 ^{1}H-NMR spectra (left) and NMR DOSY spectra (right) after 240 min of polymerization for the PMMA homopolymer (top) and the P(MMA-*co*-S) copolymer (bottom) at a polymer concentration = 10 mg ml^{-1}.

Weight-average molar masses

The NMR DOSY technique[13,24] is based on the fact that the diffusion is closely related to molecular size.[32] Indeed, in the case of spherical particles of radius R_H, the corresponding diffusion coefficient D is described by the Stokes–Einstein equation (eqn (2)).

$$D = kT/6\pi\eta R_H \tag{2}$$

where k is the Boltzmann constant, T is the absolute temperature and η is the solvent viscosity.

For a monodisperse polymer system, D can be correlated to molar mass according to eqn (3):

$$D = AM^{\alpha} \tag{3}$$

where A and α are constants specific to each polymer.

The linearization of this equation gives eqn (4):

$$\log D = \alpha \log M + \log A \tag{4}$$

To observe a linear relationship between $\log D$ and $\log M$, convection and viscosity variation must be well controlled. Since the Stokes–Einstein equation is valid for particles at

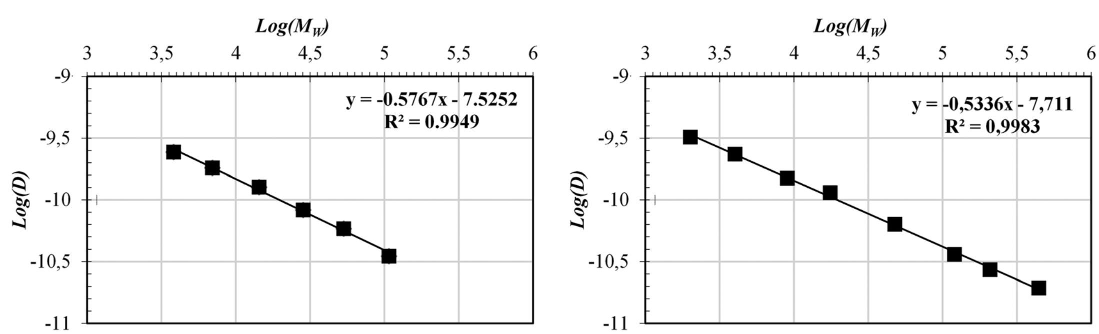


Fig. 2 PMMA (left) and PS (right) calibration curves at 25 °C in benzene for apparent M_w prediction.

least 5 times larger than the solvent molecules at infinite dilution,[24] it is highly applicable to dilute polymer solutions as described by Delsuc *et al.*[33] To enhance diffusion of the macromolecules, deuterated benzene was selected as a solvent, due to its low viscosity (0.603 cP at 25 °C) and its ability to solvate a wide variety of organic polymers.[34]

In order to apply these equations for determining the M_w of different (co)polymers and corresponding hydrodynamic radii R_H, calibration curves were first derived from NMR DOSY of a series of commercial PMMA and PS standards, and demonstrated highly linear variation of log D *vs.* log M_w (Fig. 2).

Here, we can notice the similar behavior of both standards. Only the PMMA calibration curve was applied for determining weight-average molar masses M_w of the PMMA and P(MMA-*co*-S) samples with polymerization time (Table 1 NMR DOSY).

Fig. 3 illustrates the NMR DOSY spectra of samples taken at different times for both the MMA and MMA/S polymerizations.

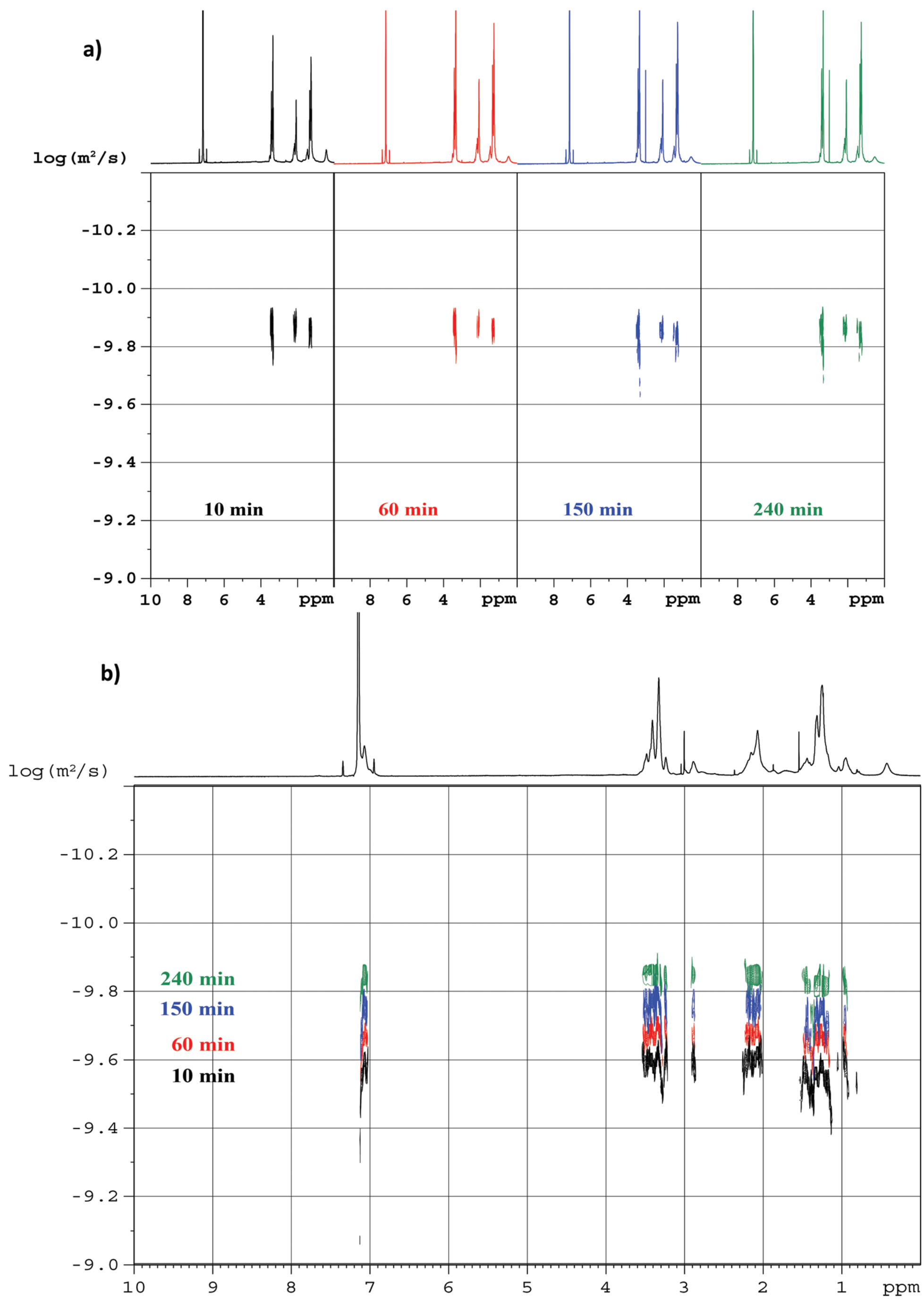


Fig. 3 NMR DOSY spectra of (a) PMMA and (b) P(MMA-*co*-S) at 10, 60, 150, and 240 min of polymerization (polymer concentration = 2 mg ml^{-1}).

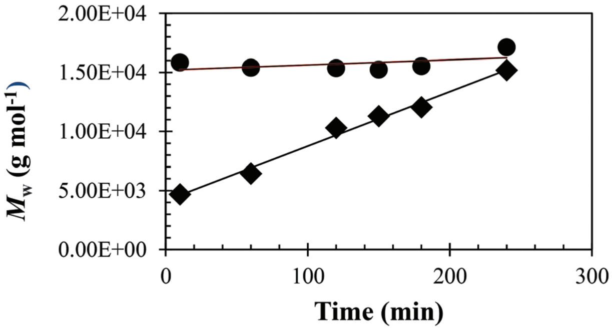


Fig. 4 $M_{w\ DOSY}$ *versus* time for PMMA (●) and (MMA-*co*-S) (◆).

The logarithmic values of D are almost constant from 10 to 240 minutes of reaction for pure PMMA (Fig. 3a) whereas they increase significantly for P(MMA-*co*-S) (Fig. 3b). These features confirm the uncontrolled behavior of the PMMA growth, with termination leading to dead chains with the same weight-average molar masses. For the MMA/S mixture, the increase of the $\log D$ values throughout the polymerization is consistent with a controlled process due to the presence of S. Note that the PMMA chain-end cannot be clearly observed in Fig. 3a due to the low polymer concentration required for precise D determination and evolution with time.

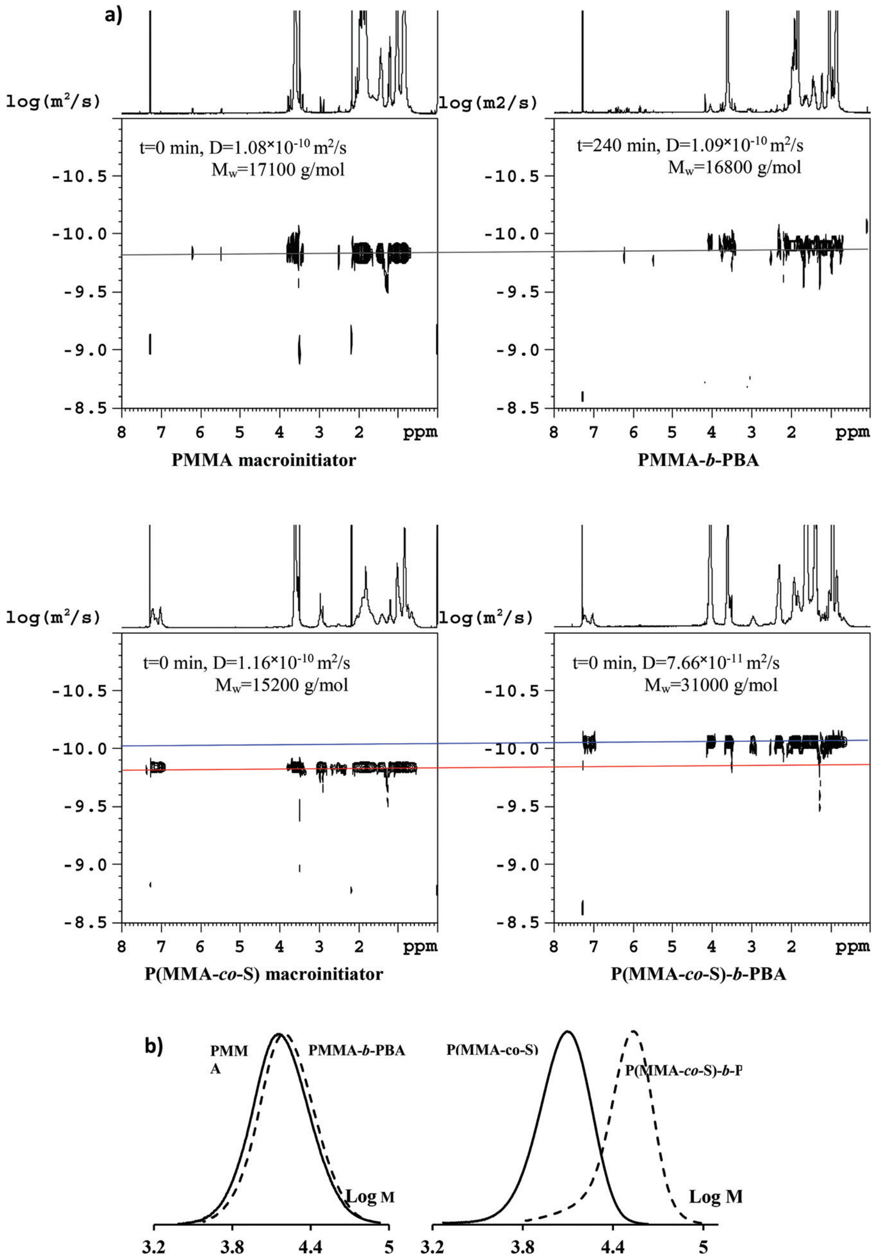


Fig. 5 (a) NMR DOSY of PMMA/PMMA-*b*-PBA (top) and P(MMA-*co*-S)/P(MMA-*co*-S)-*b*-PBA (bottom) macro-initiator/block copolymer couple and (b) overlap of PMMA (left) and P(MMA-*co*-S) (right) macro-initiator molar masses distributions with their corresponding diblock copolymers from chain extension, PMMA-*b*-PBA and P(MMA-*co*-S)-*b*-PBA respectively.

$M_{w\ DOSY}$ for all polymerization samples was then calculated using the calibration curve (Fig. 2), and values are gathered for different polymerization times in Table 1. Fig. 4 exhibits the variation of the $M_{w\ DOSY}$ *vs.* time.

The weight-average molar masses of P(MMA-*co*-S) increase linearly with time, confirming the controlled growth of the PMMA chains in the presence of S units. In contrast, PMMA molar masses are constant irrespective of the polymerization time consistent with the free radical polymerization process of the MMA in the absence of S. The evolution of $M_{w\ DOSY}$ *vs.* conversion is also reported in Fig. SI.5† and describes the same mechanism of molar masses evolution.

Chain extensions

The NMR DOSY technique was also used to verify the re-initiation efficiencies of both macro-initiators PMMA and P(MMA-*co*-S) in their chain extensions with BA. Fig. 5a illustrates the NMR DOSY spectra of the two syntheses of the diblock copolymers using PMMA and P(MMA-*co*-S) as macro-initiators.

The PMMA-*b*-PBA block copolymer at 240 min presents the same value of diffusion coefficient as the macroinitiator, reflecting the presence of only one species with constant molar mass along the polymerization and thus confirmation of lack of efficiency of the PMMA macro-initiator to extend macromolecular chains due to a large amount of dead chains. As expected, when a small percent of S is added to MMA, the NMR DOSY spectra of P(MMA-*co*-S) and P(MMA-*co*-S)-*b*-PBA show the presence of two different values of diffusion coefficient *D*, confirming the occurrence of diblock formation and consequently the controlled behavior of P(MMA-*co*-S) to grow the initial chain to reach a block copolymer. The same observations are deduced from SEC chromatograms of the PMMA-*b*-PBA and the P(MMA-*co*-S)-*b*-PBA block copolymers, resulting from chain extension of PMMA and P(MMA-*co*-S) polymers, respectively (Fig. 5b).

Extracted from the *D* values and calibration curves, the variation of $M_{w\ DOSY}$ values for each block copolymer *vs.* time and conversion is exhibited in Fig. 6 and SI-5† (right).

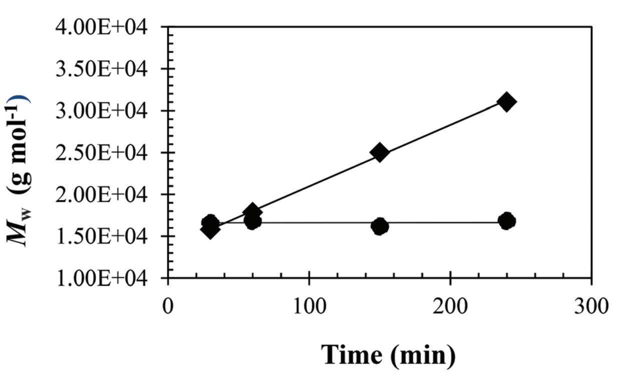


Fig. 6 $M_{w\ DOSY}$ *vs.* time of PMMA-*b*-PBA (●) and P(MMA-*co*-S)-*b*-BA (◆) diblock copolymers.

These values increase for the P(MMA-*co*-S)-*b*-PBA block copolymer, while they appear constant for the PMMA-*b*-PBA diblock copolymer, affirming one more time the presence of reversible chain ends (dissociation of the terminal alkoxyamine to create active radicals) and dead chains (irreversible termination by disproportionation due to side reactions).

Conclusion

In this work, the NMR DOSY technique was used for the first time as an efficient tool to probe a mechanism of polymerization in terms of side reactions, molar mass evolution and livingness to synthesize block copolymers. This study used nitroxide-mediated polymerization of methyl methacrylate initiated by the SG1-based alkoxyamine BlocBuilder in the presence of a small amount of styrene as a model system. The NMR DOSY analysis of a series of narrow PMMA standards demonstrated that the diffusion coefficient varies linearly with molecular weight, and a calibration curve could therefore be used to determine molar masses in subsequent NMPs. The molar masses determined by NMR DOSY closely matched those determined by SEC, but with the advantages of much faster sample preparation, shorter analysis time, and much lower solvent consumption. Furthermore, the nature of chain ends could be determined, providing a comprehensive picture of the livingness of the polymerisations. This all-in-one technique opens the door to a general study of new monomer polymerization and its controls, but efforts still have to be made to evaluate dispersity values.

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